

HW 3 – Sampling Distribution of a Difference in Sample Proportions

1) In a single town, there are two high schools: North and South (each high school has more than 2000 students in it). At North High School, the principal says that they have 18% of students arriving tardy each day. At South High School, the principal claims that they have 22% of students arriving tardy each day. The North principal plans to take an SRS of 250 students to see how many are tardy, and the South principal plans to take an SRS of 200 students to see how many are tardy. Let p_N represent the proportion of students arriving late at North and p_S represent the proportion of students arriving late at South.

(a) Describe the shape, center, and spread of the sampling distribution of $\hat{p}_N - \hat{p}_S$.

(b) The principals say that if the proportion of students tardy in South's sample is 10% or higher than the proportion of students tardy in North's sample, they will implement a new tardy policy. What is the probability of this happening? Use the sample distribution you created in part (a).

For the following, explain why you cannot use a normal sampling distribution to evaluate the probability.

2) In Wonderwood, 40% of residents own a dog. Wonderwood has a population of 1,200 residents. If a random sample of 15 residents is taken, what is the probability that the sample proportion of dog owners is less than 0.35?

3) A Babylist survey suggests that 75% of new moms are satisfied with a new bottle on the market. Babylist sends out a survey to 500 of its new mom subscribers asking their opinion on it and 155 return the survey. Of the 155 that return the survey, what is the probability that the sample proportion is between 0.70 and 0.80?

4) A recent New York Times study suggests that 30% of high school students prefer online learning. In your school, you decide to take a random sample of 300 students to see if they prefer online or in-person learning. If your school has 950 students in it, what is the probability that the sample proportion of students who prefer online learning is less than 0.25?

5) A survey indicates that 85% of Americans prefer coffee over tea. If a random sample of 50 Americans is taken, what is the probability that the sample proportion of coffee lovers is over 0.85?